KARTIK TOMAR

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EDUCATION

Dr. A.P.J. Abdul Kalam Technical University — Lucknow
Bachelor of Technology in Computer Engineering — *2023 – Present*

East Delhi Public School — Ghaziabad
Senior Secondary (Class XII) - CBSE — *2022 – 2023*

East Delhi Public School — Ghaziabad
Secondary (Class X) - CBSE — *2020 – 2021*

PROJECTS

E-Consultation Sentiment Analysis Dashboard | *React.js, Python, Flask, NLTK, Chart.js*

- **Engineered a full-stack sentiment analysis platform** for e-consultation patient feedback, classifying reviews into positive, neutral, and negative sentiments using **VADER/NLTK NLP pipelines** achieving **85%+ classification accuracy**.
- Designed an **interactive analytics dashboard** visualizing real-time sentiment trends, word clouds, and doctor/specialty-wise performance breakdowns using **Chart.js**, supporting analysis for **50+ consultations**.
- Engineered an **admin dashboard** to manage billing, menu updates, rebates, and complaints, reducing manual workload by **60%** during testing.
- Created **detailed reporting** with breakdowns by consultation category, sentiment scores, and time-based trends to assist healthcare providers in evaluating service quality and **improving patient satisfaction**.

Deep Packet Inspection & Filtering Engine | *C++17, Multi-threading, TCP/IP, Sockets, PCAP, POSIX Threads*

- **High-Throughput Protocol Parsing:** Parsed Ethernet, IPv4, TCP, and UDP headers in native C++, achieving **2.5M pps parsing rates** with minimal memory allocation overhead.
- **Multi-Threaded Concurrency Model:** Scaled worker thread pools across Load Balancers and Fast Paths using customized thread-safe queues with **std::condition_variable**, maintaining **¿98% thread utilization efficiency**.
- **Stateful Flow Tracking:** Managed **100,000+ simultaneous flows** via consistent 5-tuple hashing, ensuring 100% connection affinity for multi-threaded state management.
- **Low-Latency Rules Engine:** Extracted TLS SNI metadata from Client Hello payloads to execute flow-level IP, domain, and application blocking, dropping restricted packets with a zero memory leak rate.

ACHIEVEMENTS

- Selected as a **Finalist** in **Policy Conclave'26** at the **Indian Institute of Technology, Kanpur**.
- Ranked **Top 10** in **Hacker's Premier League** at **Guru Gobind Singh Indraprastha University (Delhi)**.
- Ranked **Top 20** in the **National AI/ML Hackathon by AVEVA**, a national-level competition organized by **IIT Hyderabad**, building AI/ML solutions as part of team **Thunders** in **Mar 2026**.
- Solved over **100+ coding problems** across platforms including **LeetCode**.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript

Web Technologies: HTML5, CSS3, Tailwind CSS, React.js, Node.js, Express.js, MongoDB, Git/GitHub

Coursework: Data Structures, Algorithms, Object-Oriented Programming, Database Management System, Operating Systems, Computer Networks

Soft Skills: Problem Solving, Communication, Decision Making